

Theoretical Part – Q1

Show that the composition of linear functions is a linear function. Show that the composition of affine transformations remains an affine function.

1. Let $f(x) = Ax$, $g(x) = Bx$,
then $f \circ g(x) = ABx = Cx$ for some matrix C
2. Similarly, for $f(x) = Ax + b$, $g(x) = Cx + d$,
then: $f \circ g(x) = A(Cx + d) + b = ACx + (Ad + b) = Ex + t$ for
some matrix E and some vector t

Theoretical Part - Q2

Assume a network is required to predict an angle (0-360 degrees). How will you define a prediction loss that accounts for the circularity of this quantity, i.e., the loss between 2 and 360 is not 358, but 2 (since 0 is 360..). Write your answer in a pseudo-code.

We'll define the following loss function:

$$L(\theta_r, \theta_l) = \min (|\theta_r - \theta_l|, 360 - |\theta_r - \theta_l|)$$

We first note that the abs induces a symmetric loss (since we don't care which angle is smaller/larger, just their difference). Additionally, the min function allows us to always select the acute angle rather than the obtuse.

Theoretical Part - Q3

1. $\frac{\partial}{\partial x} f(x + y, 2x, z)$

Define the following:

$$g(x, y, z) = f(x + y, 2x, z) = f(a_1, a_2, a_3)$$

With $a_1 = x + y$, $a_2 = 2x$, $a_3 = z$

Then

$$\frac{\partial g}{\partial x} = \frac{\partial f}{\partial a_1} \cdot \frac{\partial a_1}{\partial x} + \frac{\partial f}{\partial a_2} \cdot \frac{\partial a_2}{\partial x} + \frac{\partial f}{\partial a_3} \cdot \frac{\partial a_3}{\partial x}$$

We have: $\frac{\partial a_1}{\partial x} = 1$, $\frac{\partial a_2}{\partial x} = 2$, $\frac{\partial a_3}{\partial x} = 0$

Therefore:

$$\frac{\partial g}{\partial x} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \nabla f(x + y, 2x, z)$$

Where the evaluation point is $(x + y, 2x, z)$

$$2. \frac{\partial}{\partial x} f_1 \left(f_2(\dots f_n(x)) \right) = \frac{\partial}{\partial x} f_1 \circ g(x)$$

$$\text{For } g(x) = (f_2 \circ \dots \circ f_n)(x)$$

$$\text{Then } \frac{\partial}{\partial x} f_1 \circ g(x) = Df_1(g(x))Dg(x)$$

Expanding $Dg(x)$:

$$Dg(x) = D(f_2(f_3 \dots (f_n(x))) \cdot D(f_3 \circ \dots \circ f_n)(x)$$

And Going on recursively, we eventually get:

$$\begin{aligned} \frac{\partial}{\partial x} f_1 \left(f_2(\dots f_n(x)) \right) &= Df_1 \left(f_2 \dots (f_n(x)) \right) \cdot Df_2(f_3 \dots f_n(x)) \dots Df_n(x) \\ &= \prod_{i=1}^n Df_i(f_{i+1} \circ \dots \circ f_n(x)) \end{aligned}$$

$$3. \frac{\partial}{\partial x} f_1 \left(x, f_2 \left(x, \dots f_{n-1} \left(f_n(x) \right) \right) \right)$$

Rewriting this as : $g(x) = f_1(x, u)$, for $u_1 = f_2 \left(x, \dots f_{n-1} \left(f_n(x) \right) \right)$

$$\text{Then } \frac{\partial g}{\partial x} = \frac{\partial f_1}{\partial x} + \frac{\partial f_1}{\partial u_1} \cdot \frac{\partial u_1}{\partial x} = \nabla f_1(x) + \frac{\partial f_1}{\partial u} \cdot \frac{\partial u_1}{\partial x}$$

Like in the previous function, we expand recursively $\frac{\partial u}{\partial x}$:

$$\frac{\partial u_1}{\partial x} = \frac{\partial f_2}{\partial x} + \frac{\partial f_2}{\partial u_2} \cdot \frac{\partial u_2}{\partial x} = \dots$$

$$\text{Until we reach } \frac{\partial u_{n-1}}{\partial x} = \frac{\partial f_n}{\partial x}$$

So in total we get:

$$\frac{\partial g}{\partial x} = \frac{\partial f_1}{\partial x} + \frac{\partial f_1}{\partial u_1} \left(\frac{\partial f_2}{\partial x} + \frac{\partial f_2}{\partial u_2} \left(\frac{\partial f_3}{\partial x} + \frac{\partial f_3}{\partial u_3} \left(\dots \frac{\partial f_{n-1}}{\partial x} + \frac{\partial f_{n-1}}{\partial u_{n-1}} \cdot \frac{\partial f_n}{\partial x} \right) \right) \right)$$

$$4. \frac{\partial}{\partial x} f \left(x + g(x + h(x)) \right)$$

Defining:

$$u_1 = x + h(x), \quad u_2 = x + g(u_1)$$

$$R(x) = f(u_2)$$

We get:

$$\frac{\partial R}{\partial x} = f'(u_2) \cdot \frac{\partial u_2}{\partial x}$$

$$\text{Expanding } \frac{\partial u_2}{\partial x} = \frac{\partial}{\partial x} (x + g(u_1)) = 1 + g'(u_1) \frac{\partial u_1}{\partial x}$$

Applying the chain rule again:

$$\frac{\partial u_1}{\partial x} = 1 + h'(x)$$

So overall we got:

$$\frac{\partial R}{\partial x} = f' \left(x + g(x + h(x)) \right) \left(1 + g'(x + h(x)) \right) (1 + h'(x))$$